The Sorghum Lipid Database (SoLD)

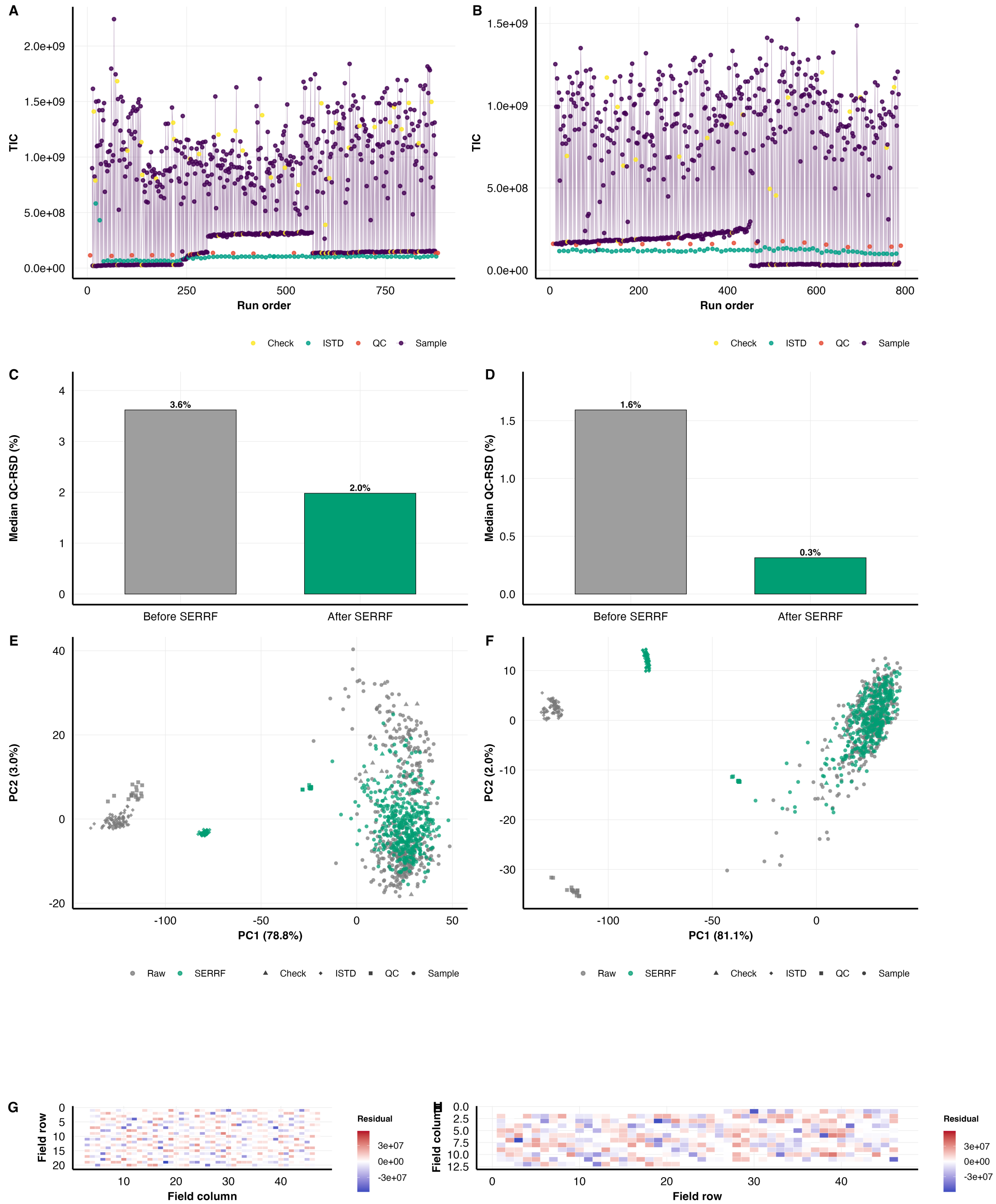
Summary of Results and Discussion

Design. The Sorghum Association Panel grown in two field trials. **CTL** is 2019, nutrient-sufficient, standard planting date. **LIN** is 2022, reduced nitrogen and phosphorus, earlier planting to impose cold at establishment, and no insecticide, herbicide or pesticide. 394 accessions in CTL and 363 in LIN; 243 lipid species quantified by high-resolution LC–MS.

1. Results

1.1 QC and normalization

- **Method.** SERRF normalization on the vendor peak tables, then SpATS spatial correction for field position, with model residuals added back to the fitted mean so genotype signal is preserved. QC-RSD compared before and after SERRF, plus run-order drift, injection-level PCA, and maps of SpATS residuals across the field.
- **Found.** Analytical performance is comparable between trials, so CTL–LIN differences downstream are not instrument drift. Spatial correction removes plot-to-plot field variation that would otherwise look like genotype effects.



Supp. Fig. S2. QC diagnostics – run order, SERRF normalization, injection PCA, and SpATS spatial correction.

1.2 Race and population structure explain almost nothing

- **Method.** Kruskal–Wallis with $\varepsilon^2 = (H - k + 1)/(n - k)$ as effect size, truncated at zero and BH-corrected. Two groupings tested – the five classical botanical races plus a Mixed group, and six marker-based genetic clusters. Plus a PCA of the 13-class composition per trial on $\log_{10}$ values scaled to unit variance.

- **Race explains nothing.** No trait significant in either trial after BH correction (smallest $q = 0.468$; effects $\varepsilon^2 \leq 0.027$).
- **Genetic cluster explains one thing.** PS under LIN ($q = 1.11 \times 10^{-6}$, $\varepsilon^2 = 0.104$). Under CTL the same trait gives only a weak, non-significant trend.
- **Ordination agrees.** Accessions of one race or cluster are dispersed rather than grouped; race or cluster explains at most 1.6% of variance.
- **Why it matters.** The lipidome is not a readout of population history, so lipid GWAS in this panel is not re-discovering structure.

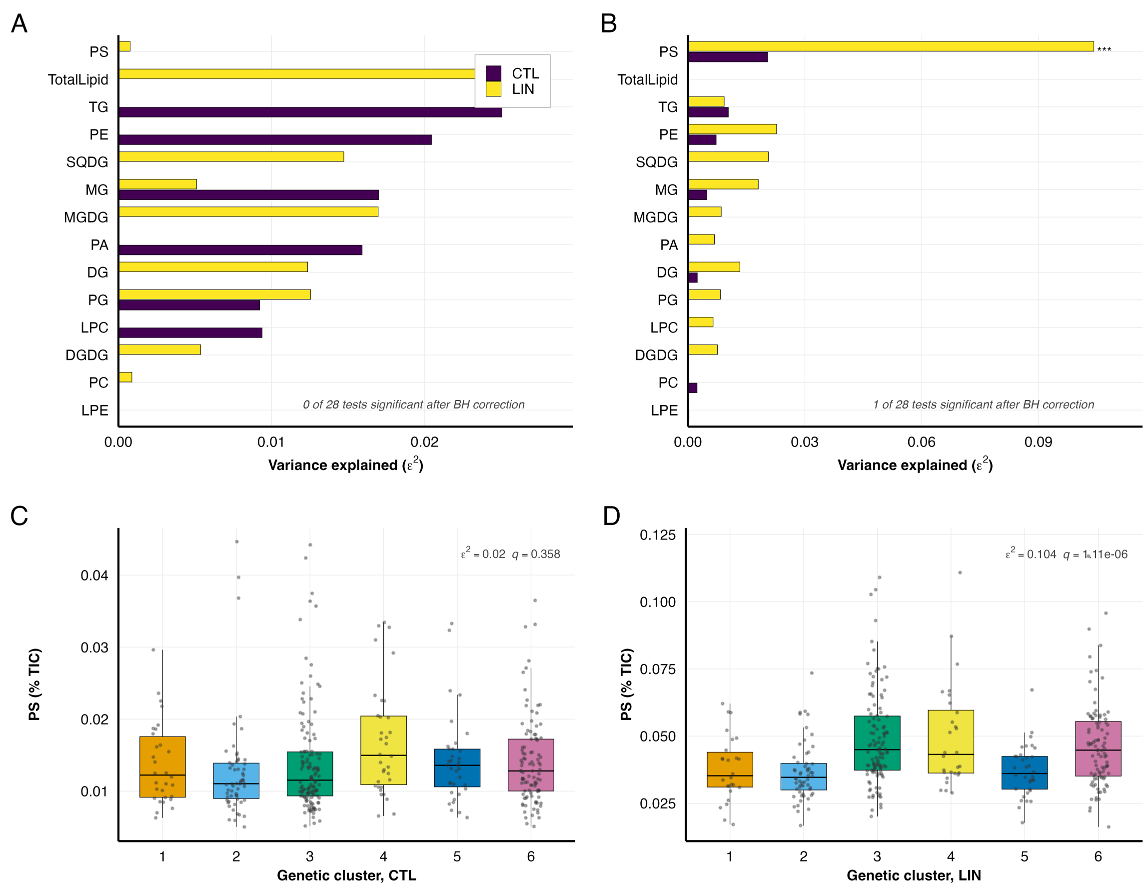
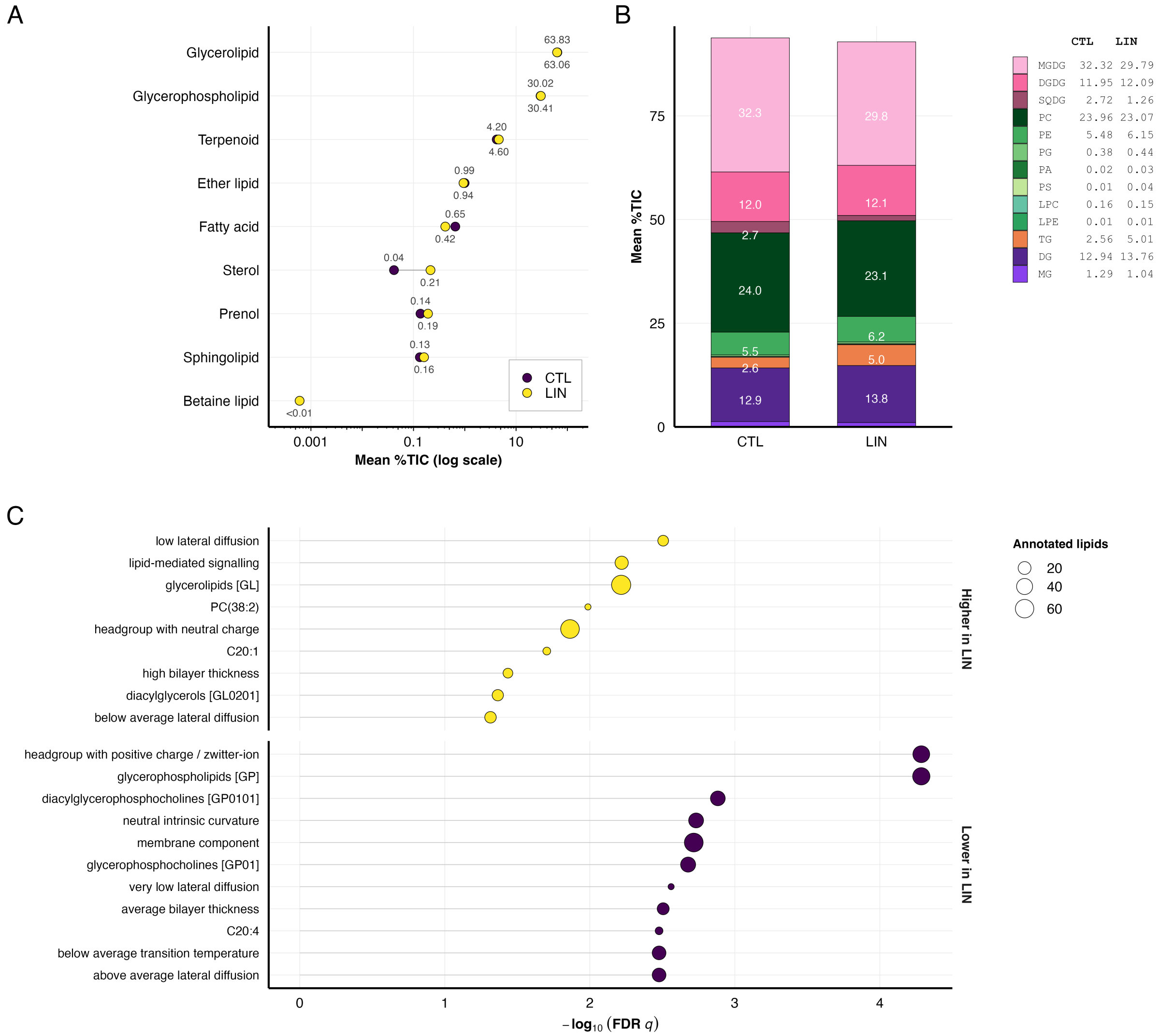


Figure 1. Botanical race and marker-based cluster explain little lipid-class variation; PS under LIN is the single exception.

1.3 Lipidome overview and the four compositional axes

- **Method.** Species inventory; composition as %TIC taken per sample and then averaged, so a few high-signal samples cannot dominate; species-level PCA within each trial; LION ontology enrichment as the one direct CTL–LIN contrast.
- **Inventory.** 243 species – 152 in both trials, 49 CTL-only, 42 LIN-only, where trial-specific means below detection threshold rather than confirmed absent.
- **Composition.** MGDG 32.3 → 29.8%, PC 24.0 → 23.1%, DGDG 12.0 → 12.1%, DG 12.9 → 13.8%, **TG 2.6 → 5.0%**, **SQDG 2.7 → 1.3%**. The 13 focal classes are ≈93% of annotated signal.
- **Minor pools.** Sterols show the largest *proportional* change of any superclass, 0.042 → 0.215%, almost all β-sitosterol at 4.6-fold.
- **Structure.** Lipid class strongly structures species-level variation; TG forms a distinct cluster away from membrane lipids in both trials.
- **LION.** A head-group charge contrast, with storage and lipid-droplet terms up under LIN and glycerophospholipid and membrane-component terms down.
- **The four axes, defined here.** (i) low SQDG, (ii) high TG, (iii) a PS-centred phospholipid shift, (iv) an LPC/LPE balance shifted toward LPE (LPC flat, LPE up 1.4-fold, ratio 17.0 → 11.3).



	CTL	LIN
MGDG	32.32	29.79
DGDG	11.95	12.09
SQDG	2.72	1.26
PC	23.96	23.07
PE	5.48	6.15
PG	0.38	0.44
PA	0.02	0.03
PS	0.01	0.04
LPC	0.16	0.15
LPE	0.01	0.01
TG	2.56	5.01
DG	12.94	13.76
MG	1.29	1.04

Figure 2. Lipid-class composition. (A) every annotated superclass on a log scale; (B) the 13 focal classes; (C) LION ontology terms, the only direct CTL–LIN contrast in the figure.

1.4 GWAS

- **Method.** GEMMA univariate mixed linear model, centered kinship from SNP genotypes, first three genotype PCs as covariates, MAF 0.05 and heterozygosity 0.8 filters, Bonferroni threshold $p \leq 8.19 \times 10^{-9}$. Run for every individual species, every class sum and every class ratio, in both trials.
- **Found.** Association signal is concentrated under CTL and diffuse under LIN – fewer, sharper peaks in the control trial against a broader, more distributed architecture in low-input.

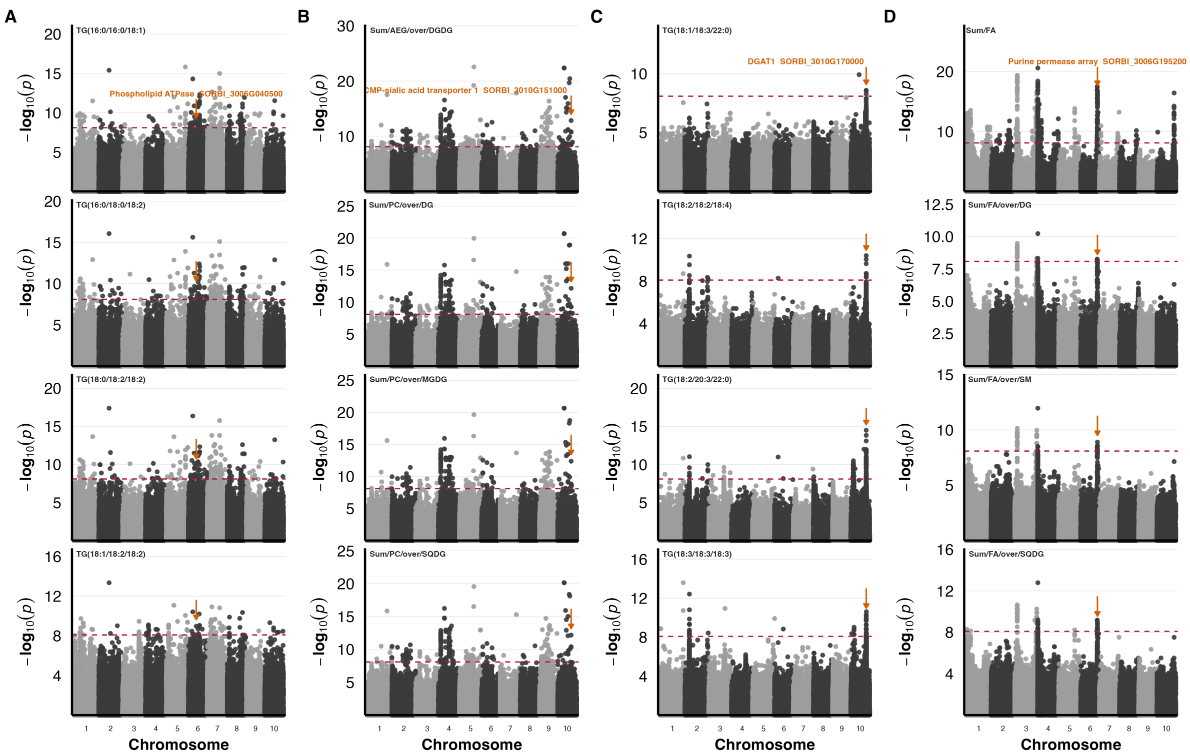


Figure 3. Manhattan plots for representative traits in each trial and trait layer. Arrows mark the candidate loci discussed in the text.

1.5 GO enrichment – the principal methods contribution

This is the part worth reading in full. Four steps:

1. **Stratified testing.** GO over-representation is run *separately* within each lipid class × trial × trait layer × ontology. Stratifying by class is what makes a term interpretable, since it attributes an enrichment to the traits of one class rather than to the candidate set as a whole.

2. **LD-aware permutation null.** In sorghum a significant SNP tags every gene in its LD window rather than one gene, so a naive GO test counts the same association repeatedly and inflates significance. Every term was therefore tested against a null built from 8,000 random draws of **gene-density-matched 250 kb intervals**. Density matching matters because gene-dense regions contribute more genes and would otherwise appear enriched for everything. This yields q_{LD}, corrected for mapping structure rather than only for multiple testing.

3. **Two-pass collapse.** Two things inflate the term list. The Gene Ontology nests parent and child terms, so one gene set surfaces as several terms; and class sums and ratios share candidate genes by

construction, so one gene set recurs across classes, layers and both ontologies. Terms with *identical* candidate-gene sets were merged, first within a stratum and then across strata.

4. **Interpretability filter.** For the main figure, only loci whose GO term draws on at most 30 background genes. This is a criterion of specificity, not of significance – *zinc ion binding* (1,015 background genes) passes the permutation null while naming no biological process. Interval count was deliberately *not* used to select terms, because it scales with GO term size and would discriminate against precisely the small, specific terms of interest (terms in three or more intervals average 175 background genes at 10-fold enrichment; those below average 24 genes at 23-fold).

- **Numbers.** 103 enrichments → 101 pass the LD-aware null → collapse to **50 distinct loci** (17 CTL, 33 LIN) → 19 shown in Figure 4.
- **The asymmetry is real, not a filter artefact.** 17 CTL loci against 33 LIN, from roughly half as many candidate genes, and all but one CTL survivor is a broad category.
- **LIN names processes.** Nitrate assimilation and transport 31-fold, cellular response to phosphate starvation 17.4-fold, sphingolipid metabolic process 32.5-fold, jasmonate signalling / defense / wounding 34.5-fold.
- **CTL does not.** Zinc ion binding, proteolysis, peptidase activity, transmembrane transport.
- **Two failure modes illustrated.** The purine permeases are specific but reduce to one tandem array recovered 25 times; zinc ion binding is spread over six to eight intervals but names nothing.

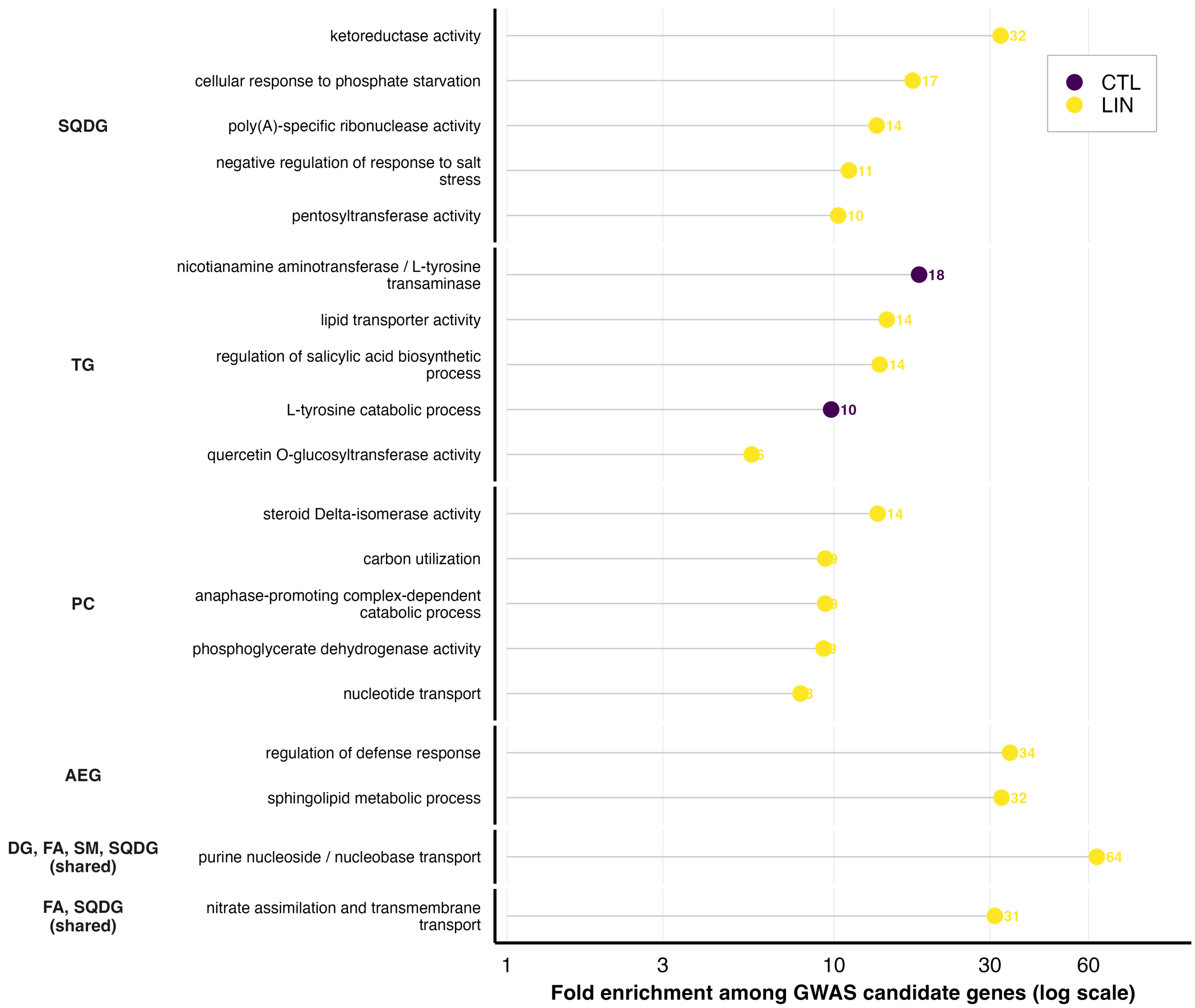
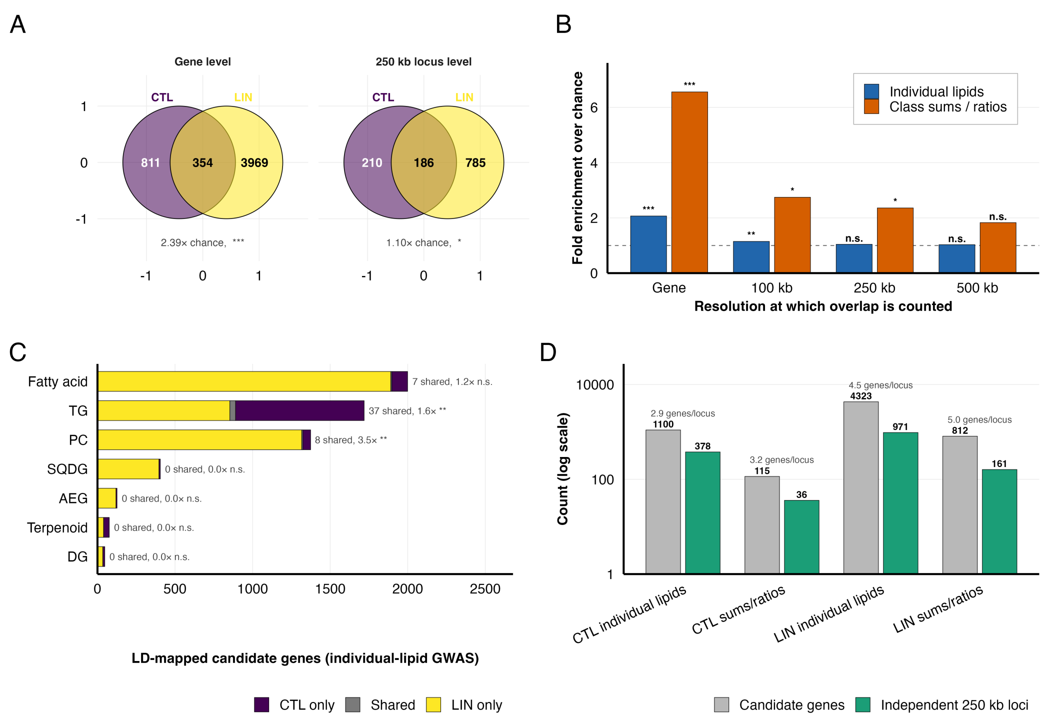


Figure 4. GO terms enriched among LD-mapped candidate genes, grouped by the lipid class whose traits produced them. Nineteen loci shown after collapse and the 30-background-gene filter.

1.6 Cross-trial overlap collapses to chance at locus level

- **Method.** One-sided hypergeometric (Fisher right-tail) test at two resolutions. At **gene level** the background is the 34,027 annotated *S. bicolor* v3.1 gene models. At **locus level** the genome is partitioned into non-overlapping 100, 250 and 500 kb windows, with background the windows containing at least one annotated gene (4,844 / 2,279 / 1,239 respectively). Reported as fold enrichment over the expected $|A||B|/N$ plus the Jaccard index, which is insensitive to the large size difference between the CTL and LIN sets.
- **Gene level looks impressive.** 354 genes shared against 148 expected, 2.39×, significant in both trait layers.
- **It decays to chance once linkage is collapsed.** 1.04× at 250 kb for individual lipids. Loci recovered in the two trials are no more concordant than two independent samples of the same genome.
- **Sharing a gene rarely means sharing a phenotype.** Only 25.4% of the 354 shared genes were associated with a lipid class in common, falling to 18.0% in the individual-lipid layer.
- **LD inflates counts three- to five-fold.** CTL 1,100 genes → 378 loci; LIN 4,323 → 971.
- **Class composites are the exception.** They retain a significant cross-trial excess at 100 and 250 kb where individual species do not, because aggregation averages over acyl-chain heterogeneity and measurement noise.
- **Take-home.** Cross-environment comparisons must be made at the locus level, not the gene level.



Supp. Fig. S8. Cross-trial candidate overlap at gene and locus resolution, and its decay with window size.

1.7 LINEX reaction mapping and reaction-balance scoring

- **Method.** LINEX2 reconstructs annotation-guided lipid reaction networks from curated rules (head-group interconversion, acylation/deacylation, elongation, desaturation). Sorghum is not a supported organism, so *Oryza sativa* was the reference – these are therefore hypotheses, not demonstrated flux. Four branches connecting the DG/MG, TG and lysophospholipid pools; three from LINEX2, the fourth (LCAT*) defined manually from reaction specificity and marked with an asterisk throughout.
- **Reaction-balance score.** Within one sample, $S_i = \text{mean}(\ell_{\text{products}}) - \text{mean}(\ell_{\text{substrates}})$, where $\ell_{i,c} = \log_{10}(C_{i,c} + \varepsilon_i)$ on class sums. Being a within-sample log-ratio, a uniform multiplicative difference between the trials cancels – exactly so in the absence of the pseudocount, and to a very good approximation with it. What it does *not* remove is an effect acting differentially on lipid classes, because a branch's products and substrates are different classes. Compared with paired, genotype-matched Wilcoxon tests across the 355 shared genotypes, which controls for genotype but not for year, planting date or acquisition batch. The direction is therefore a description of the two trials, not evidence that management caused it.
- **Found.** All four branches shift strongly and consistently – positive for the three TG-producing branches (PNPLA3 +0.327, LCAT* +0.238, LRO1 +0.232) and negative for the hydrolytic TG→DG branch (PNPLA1 −0.268), BH-adjusted $p < 10^{-56}$ across 355 genotype pairs. This recapitulates the lipidome-wide picture at the level of a small set of candidate reactions rather than as independent class shifts.

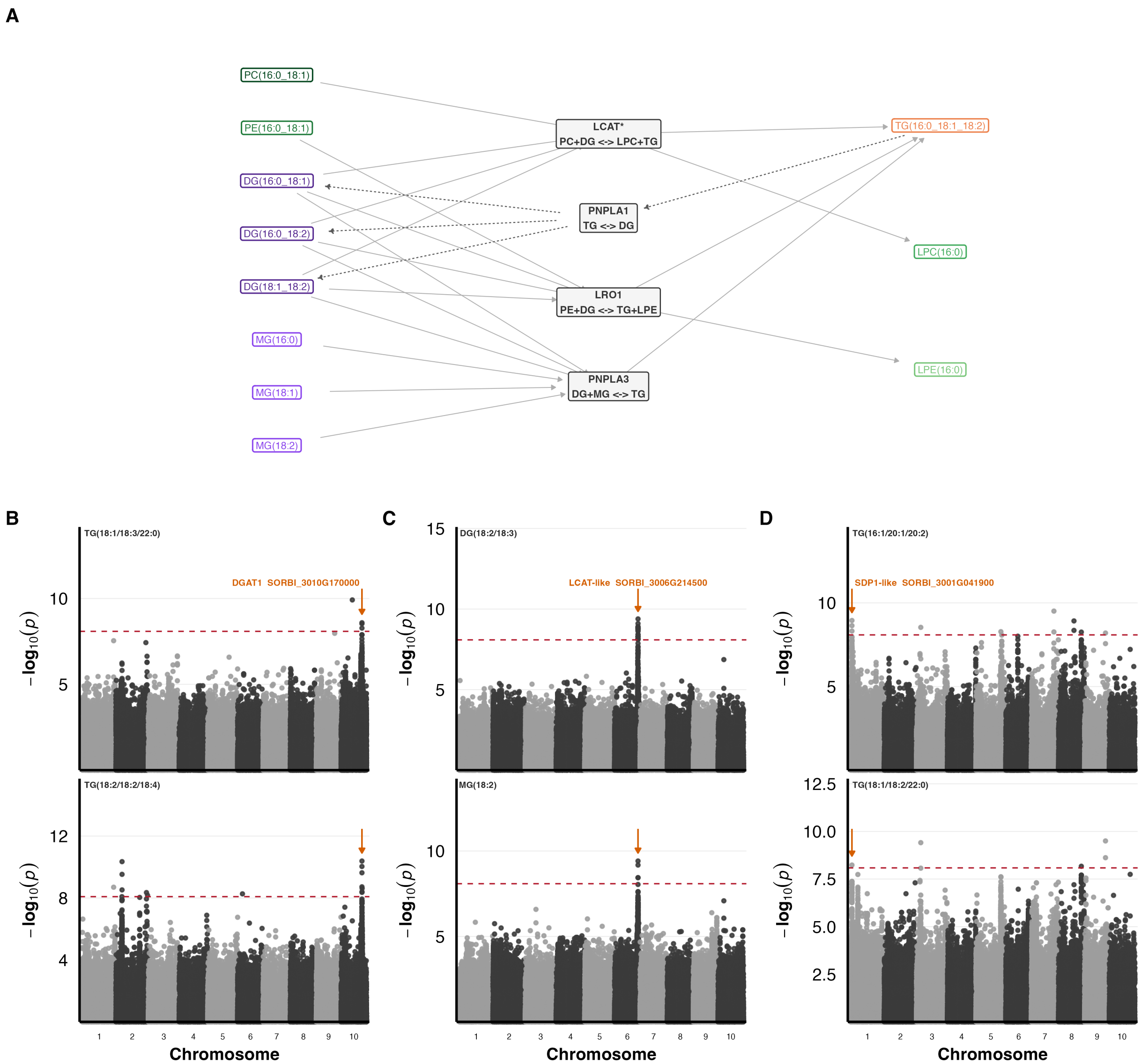


Figure 5. LINEX reaction network, branch-level reaction balances, and GWAS support for the branch-supporting candidate genes.

1.8 Interactive SoLD application

- Fourteen modules – data preview, boxplots, correlation, PCA, t-SNE/UMAP, heatmap, GWAS, gene hits, genotype, Venn, volcano and network – each with dataset and data-type (raw, %TIC, CLR) selectors.
- Worked example moves from a terpenoid class view to lupeol to a terpene cyclase annotated for triterpenoid biosynthesis and localised to the lipid droplet, demonstrating that classes outside the 13 focal ones remain reachable.

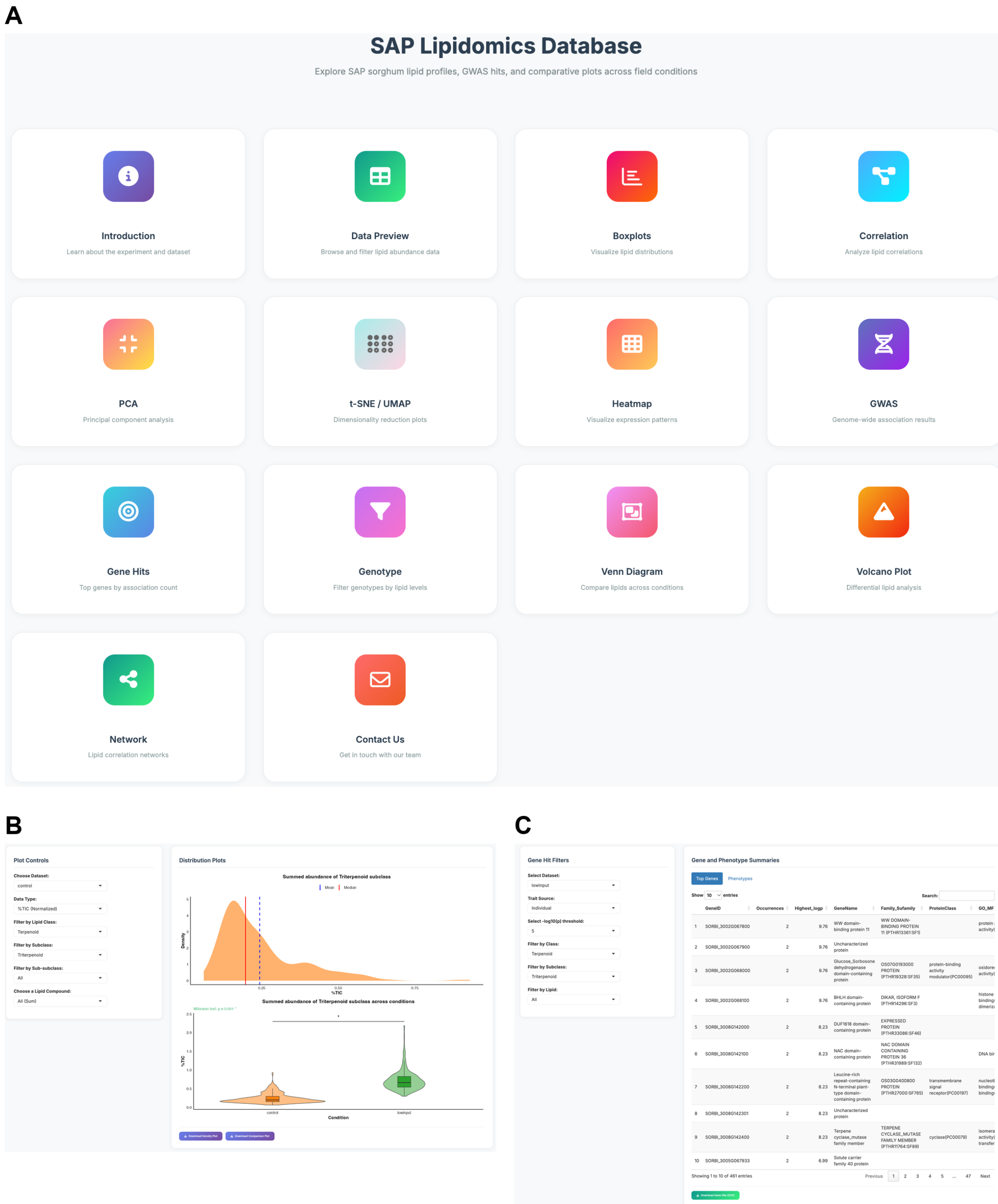


Figure 6. The SoLD Shiny application, showing the class-level boxplot view and the gene-hit table for the same phenotype.

2. Discussion

Two caveats framing everything below. Gene functions are assigned by orthology to characterised genes in other species, so they say what the family can do, not what the sorghum locus does in a field. And because candidates come from LD blocks, most regions hold several genes – an enrichment resting on fewer than three independent 250 kb intervals is one locus, not a pathway.

2.1 Integrated interpretation across the two trials

Why the SQDG decline is the most informative result

- **The expectation it violates.** Under phosphate limitation plants replace phospholipids with non-phosphorus surrogates, so SQDG and DGDG should *rise*. LIN had reduced P, and SQDG *fell*. Either the plants were not experiencing phosphate starvation in the physiological sense, or something else overrode the replacement response.
- **The phosphate machinery was genetically active.** `SORBI_3001G028500`, an SPX-domain phosphate-status signalling protein of the OsSPX4 type, is a LIN candidate enriched under *cellular response to phosphate starvation*, and it associates with FA/SQDG and SQDG/FA *ratios* – i.e. with the sulfolipid pool specifically. So phosphate status is being sensed and it is genetically linked to SQDG; the pool just moves the wrong way.
- **Hypothesis 1 – sulfur, not phosphorus, is the limiting input.** SQDG is a plastid lipid whose sulfoquinovose head group is built from sulfate imported into the chloroplast and reduced there. `SORBI_3006G244000` is a SULTR3-family sulfate transporter; in Arabidopsis SULTR3 proteins sit in the chloroplast envelope and carry plastid sulfate uptake. LIN received no fertiliser and no pesticide, so sulfate supply was not topped up either. If plastidic sulfate is short, the plant cannot execute the phospholipid-to-sulfolipid swap even while sensing low P – the substrate for the head group is missing. This is the cleanest explanation the data support, and it is testable.
- **Hypothesis 2 – the sulfolipid synthases themselves.** Two SQD2-like sulfoquinovosyltransferase loci are LIN candidates (`SORBI_3003G076900`, `SORBI_3002G000600`), the enzyme that transfers sulfoquinovose onto DAG. Both were recovered through fatty-acid phenotypes rather than SQDG levels, so this is suggestive rather than established, but it puts allelic variation at the committed step of sulfolipid synthesis inside the LIN candidate set.
- **Hypothesis 3 – less thylakoid to put it in.** SQDG is a thylakoid membrane lipid. LIN was planted early to impose cold, and cold plus low N both reduce photosynthetic membrane per unit tissue. A smaller plastid membrane compartment lowers SQDG regardless of phosphate status. MGDG and DGDG behaviour is consistent with a plastid-membrane component to the shift.
- **Hypothesis 4 – severity and stage.** Classical replacement is described under severe or complete P withdrawal, usually hydroponic. LIN was *reduced* P in field soil with a residual pool, and sampled at a single developmental stage that differed from CTL. A partial P deficit may simply never cross the threshold that triggers membrane lipid replacement.
- **The phosphate transporter points the same way.** `SORBI_3002G116100`, a PHT1-family H^+/Pi cotransporter whose closest characterised rice relative is OsPT4, is a LIN candidate for five species at $p = 1.1\times10^{-9}$ ($r^2 = 0.87$) – and four of the five are *neutral storage lipids*, including TG(18:3_18:3_18:3), not phospholipids. A phosphate transporter reporting on triacylglycerol is the opposite of what phospholipid-to-glycolipid replacement predicts. It is the weakest association discussed and only marginally clears Bonferroni, so it is a pointer, not a pillar.
- **What we can and cannot claim.** We can say the LIN lipidome is not the textbook phosphate-starvation lipidome, and that the candidate genes put sulfur supply and sulfolipid synthesis on the table as the alternative. We cannot attribute the decline to any single factor, because nutrient level,

temperature, planting date, developmental stage and year are confounded in this design.

The rest of the integrated picture

- TG accumulation comes with reorganised DG–TG coordination rather than a uniform rise in neutral lipids – ratios involving TG, DG, PE and MGDG all shift toward TG, the PS–TG correlation changes sign, and the LINEX balances split by branch direction (three TG-producing branches positive, the hydrolytic TG→DG branch negative).
- PS rose relative to other phospholipids and its correlation with PC reversed, which is head-group redistribution inside a reorganised network rather than membrane expansion. PE gained the most unsaturation of any membrane class. But PS and the lysophospholipids are each under 0.2% of TIC against tens of percent for MGDG, so these are redistributions within small pools, and compositional transforms exaggerate them. PS is also the only trait clearly structured by genetic cluster, and only under LIN.
- Genetic architecture is strongly environment-dependent – after collapsing linkage blocks, CTL and LIN loci overlap no more than chance (1.04× at 250 kb) and only ∼25% of shared genes map to the same lipid class. Class-level composites are the exception and are the more robust unit for cross-environment comparison. Genotype-by-nutrient cannot be separated from genotype-by-year or genotype-by-management here.

2.2 A TIFY/JAZ cluster recovered in both trials

- Four TIFY paralogues in a 290 kb stretch on chromosome 1 – `SORBI_3001G259300`, `SORBI_3001G259400` and `SORBI_3001G259600` – **recovered under both regimes but reporting different lipids**. Under CTL, `SORBI_3001G259400` and `SORBI_3001G259600` associate with nine 16:0-containing storage triacylglycerols, e.g. TG(16:0_18:1_18:1) and TG(16:0_18:1_20:1). Under LIN those two plus `SORBI_3001G259300` instead associate with AEG(o-15:2_15:0), PC(14:0_18:3) and 11,14,17-eicosatrienoic acid, with the top signal strengthening to $p = 1.6 \times 10^{-12}$.
- **Mechanistically tight rather than coincidental.** JAZ/TIFY proteins are the repressors of jasmonate signalling, and jasmonate is made from α-linolenic acid (18:3) released from plastid galactolipids. The LIN traits carry exactly that chemistry – an 18:3 phosphatidylcholine and a free C20:3 fatty acid.
- In the AEG test the cluster maps to a collapsed locus for *regulation of defense response*, *regulation of jasmonic acid mediated signaling pathway* and *response to wounding*, enriched 34.5-fold over sixteen background genes ($q_{\mathrm{LD}} = 0.0068$) – and unlike the other tandem loci, on *two* independent 250 kb intervals, not one.
- Cold support in three systems – rice *OsTIFY* (cold-induced, ABA-independent), pepper *CaTIFY7*/*CaTIFY10b* (silencing confers cold sensitivity, overexpression improves tolerance) and coordinated cold induction in sweet orange. LIN was planted early to impose cold, so this fits.
- **Caveats.** LD peaks at $r^2 = 0.88$, so the causal variant could be any of the four paralogues or a neighbour; trial-specific associations do not prove functional switching, since only genome-wide-significant traits are counted; and three genes against sixteen background makes the fold estimate imprecise.

2.3 Under CTL, membrane organization and amino-acid metabolism

- **The membrane link.** `SORBI_3006G040500`, a phospholipid-transporting P4-type ATPase, associated with 27 TG phenotypes. P4-ATPases are flippases that move lipids between membrane leaflets; the Arabidopsis relative ALA2 is strongly PS-specific. It contributes to no enriched term and is carried on annotation alone.

- **The tyrosine locus is the more interesting half.** The enriched term is *L-tyrosine catabolic process*, four genes on three chromosomes, and *every* phenotype at those loci is a triacylglycerol. Tyrosine degradation runs through tyrosine aminotransferase and 4-hydroxyphenylpyruvate dioxygenase to homogentisate, then to fumarate and **acetoacetate, which feeds acetyl-CoA** – the substrate for fatty-acid synthesis.
- `SORBI_3005G152200`, a chromosome 5 aminotransferase, alone explains the nine-species 16:0 TAG block ($p = 9.7 \times 10^{-13}$). `SORBI_3002G104200`, annotated with HPPD activity, places the next step of the same pathway at an unlinked locus.
- **The same amino acid is committed to dhurrin under LIN**, so tyrosine is a single metabolic branch point that the two trials resolve differently – catabolism to lipid carbon under CTL, cyanogenic glucoside synthesis under LIN.
- **Caveat.** Two adjacent chromosome 2 genes fall in the nicotianamine aminotransferase subfamily (iron-uptake phytosiderophores in grasses), and the same three genes are enriched for both the tyrosine transaminase and the nicotianamine transaminase terms – annotation ambiguity within one subfamily. α-Tocopherol, the other homogentisate route, maps 1.1 Mb away, so the two are separable here.

2.4 Under LIN, nitrogen candidates

- **Uptake.** Three nitrate transporters from two families – `SORBI_3004G009400` (NRT2.1-like, high-affinity; drives nitrate transport/assimilation enrichment in both the FA and SQDG sets), `SORBI_3008G081400` (NPF/NRT1; the strongest LIN single-lipid association in the panel) and `SORBI_3001G133900`, whose rice orthologue OsNPF2.4 handles low-affinity root-to-shoot nitrate transfer.
- **Storage rather than uptake.** A chromosome 1 locus overlapping the dhurrin cluster links seven candidates to FA(22:1), 11,14,17-eicosatrienoic acid, TG(14:0_18:3_18:3) and twelve of the thirteen PA-to-class ratios. Four of the seven cover consecutive steps of dhurrin metabolism – a CYP71E1-reaction monooxygenase (`SORBI_3001G012000`, $r^2 = 1.00$), a glycosyltransferase (`SORBI_3001G012400`), a MATE-family vacuolar transporter (`SORBI_3001G012600`, $r^2 = 1.00$) and a glutathione transferase (`SORBI_3001G012500`).
- **The logic.** Nitrogen induces dhurrin synthesis in older sorghum, so under reduced N less tyrosine-derived carbon and nitrogen is committed to dhurrin, freeing precursor for lipid synthesis. Consistent with the associations being *ratios* rather than individual species. Dhurrin itself was not measured, so this rests on annotation inside one LD block.

2.5 Under LIN, phosphorus and sulfur converge on SQDG

- The three genes are `SORBI_3001G028500` (SPX phosphate-status signalling), `SORBI_3006G244000` (SULTR3 plastid sulfate transporter) and `SORBI_3002G116100` (PHT1 phosphate transporter) – see the SQDG discussion above, where each is used.
- **The one-line argument.** Phosphate limitation alone should *increase* SQDG; SQDG was depleted; a sulfolipid needs sulfur, and a plastid sulfate transporter is sitting in the candidate set. Restricted plastidic sulfate supply is the plausible alternative hypothesis, not a demonstrated mechanism.

2.6 Under LIN, a cold-induced MADS regulator

- `SORBI_3003G406800`, a MADS-box transcription factor and the **strongest LIN fatty-acid association in the panel** ($p = 2.7 \times 10^{-17}$), lead SNP intragenic ($r^2 = 1$), signal recurring across FA(22:1), PC(18:2_20:4), total FA and multiple FA-to-class ratios.
- Cold-induced in sorghum within 6 h ($\log_2$FC 1.04, adj. $p = 5.7 \times 10^{-19}$) and a hub of the early shock-phase co-expression module ($k_{\mathrm{ME}} = 0.91$).

- Its closest annotated maize homologue at 84% identity is *ZmMADS69*, the flowering-time QTL acting through ZmRap2.7 to relieve repression of the florigen *ZCN8*, and a target of tropical-to-temperate selection.
- LIN was planted early specifically to impose cold, and cold acclimation proceeds partly through acyl-chain remodelling, which makes fatty-acid traits the expected point of contact. The association may reflect direct regulation of lipid metabolism or a shift in developmental stage at sampling; this design cannot separate them.

2.7 Reaction-supported remodeling candidates under LIN

- **DG input.** `SORBI_3003G107900`, non-specific phospholipase C6, associates with PC(18:2_20:4), a plausible substrate – **but linkage cannot separate it from five sterol C-8 isomerase (EXPERA) genes in the same 137 kb window on chromosome 3**. All six share one association with the same species at $r^2 \geq 0.95$ and together form the most specifically annotated locus in the enrichment set (13.6-fold over six background genes).
- **Branch (i), PC+DG↔LPC+TG.** Two LCAT-superfamily candidates – `SORBI_3006G214500`, which links to the branch substrates DG(18:2_18:3) and MG(18:2), and `SORBI_3004G341900`, annotated as a phospholipid–sterol O-acyltransferase. The second allows an alternative chemistry: sterol esterification using PC as acyl donor would release LPC *without* producing TG. That fits LIN sterol demand, where β-sitosterol rises more than four-fold – the largest proportional superclass change – and it dovetails with the EXPERA cluster above.
- **TG synthesis.** *DGAT1* (`SORBI_3010G170000`) is the clearest genetic link to the TG phenotype, sitting under a repeatedly mapped chromosome 10 crude-fat QTL, with five of six associated phenotypes polyunsaturated triacylglycerols including TG(18:3_18:3_18:3).
- **TG breakdown.** `SORBI_3001G041900`, an *SDP1*-like patatin lipase clustering with *SbSDP1-L*, associates with three TG species and, at class level, with total TG and the TG/SQDG ratios – a bulk-pool lipase rather than a species-specific one.
- **A sphingolipid module outside the DG–TG branches.** Three genes spanning three consecutive pathway steps on three chromosomes – a serine C-palmitoyltransferase (`SORBI_3001G434900`), a DAGK-domain kinase (`SORBI_3007G041500`) and a ceramide glucosyltransferase (`SORBI_3003G394100`) – all associated with AEG(o-15:2_15:0), the only ether-linked species that increases under LIN. Supported by three independent intervals. This is genetic evidence, not abundance evidence: sphingolipid levels barely change and the species are trace. Still the most complete candidate pathway in the study and the most direct target for functional follow-up.

What is novel methodologically. The LD-aware permutation null and the two-pass gene-set collapse for GO enrichment, and the demonstration that gene-level GWAS overlap between environments is largely a linkage artefact that disappears at locus level.